

Hybrid DNS and Route 53 at Scale

Two resolvers that know nothing about each other, and how to introduce them.

Your VPC has a resolver. Your data centre has a resolver. Neither knows the other exists until you **introduce them, in each direction separately.**

How a VPC resolves a name

01

Instance asks

The query goes to the VPC resolver at base plus two.

02

Private zone

Any private hosted zone associated with this VPC is checked.

03

Resolver rule

A forwarding rule may send the query to on-premises.

04

Public DNS

Otherwise it resolves publicly, as normal.

ON THE EXAM

A private hosted zone only answers for VPCs it is associated with. Association is the step people forget.

Inbound or outbound endpoint?

INBOUND ENDPOINT

On-premises asks AWS

- Gives on-prem an IP to query
- Resolves private hosted zone names
- Point your data centre DNS at it
- Traffic comes in over DX or VPN

OUTBOUND ENDPOINT

AWS asks on-premises

- Lets the VPC resolver forward out
- Paired with a forwarding rule
- The rule names the domain to forward
- Traffic goes out over DX or VPN

THE TIEBREAKER

Name the direction of the question, not the traffic. Who needs an answer decides which endpoint you need.

Resolver rules and sharing them

- ◆ A forwarding rule says: for this domain, ask these servers. It needs an outbound endpoint.
- ◆ Rules are **shareable with AWS RAM**, so one networking account can define them once for the organisation.
- ◆ A rule must still be **associated with each VPC** that should use it. Sharing alone changes nothing.
- ◆ The most specific matching domain wins, in the same spirit as longest prefix match.

ON THE EXAM

Sharing and associating are two steps. A shared rule that nobody associated is the commonest hybrid DNS failure.

Private hosted zones

- ◆ **Answers only for associated VPCs.** Association can cross accounts, and is authorised through the API.
- ◆ **Split-horizon** works: a public and a private zone for the same name, answering differently by where you ask from.
- ◆ The VPC needs **DNS hostnames and DNS resolution enabled**, or the zone is silently ignored.
- ◆ In a shared VPC, the **owner** associates the zone. Participants cannot.

ON THE EXAM

Split-horizon is how one hostname reaches a private IP internally and a public endpoint externally, with no client changes.

Route 53 Profiles bundle private zones, Resolver rules and firewall groups into **one shareable object**.

The routing policies

POLICY	ROUTES BY	CLASSIC USE
Simple	One answer, no logic	A single endpoint
Weighted	Proportions you set	Canary releases, A/B splits
Latency	Lowest measured latency	Fastest experience globally
Failover	Health of the primary	Active-passive disaster recovery
Geolocation	Where the user is	Data residency, language, licensing
Geoproximity	Distance, with a bias dial	Shifting load between Regions

ON THE EXAM

Multivalued answer also exists: several healthy IPs returned at random. It is not a load balancer, but it is health-aware.

Latency or geolocation?

LATENCY

Fastest, wherever that is

- Optimises the user experience
- Which Region is fastest can change
- No guarantee about jurisdiction
- Use when speed is the goal

GEOLOCATION

Deterministic by location

- Guarantees which endpoint answers
- Continent, country or US state
- Set a default for unmatched queries
- Use when the rule is legal

THE TIEBREAKER

Compliance and data residency mean geolocation. Latency optimises speed and promises nothing about where data lands.

Health checks, and the private resource trap

- ◆ **Endpoint checks** probe an IP or domain. **Calculated checks** combine other checks. **CloudWatch alarm checks** follow a metric.
- ◆ **Route 53 health checkers live on the public internet.** They cannot reach a private IP address.
- ◆ So for a private resource, publish a metric and use a **CloudWatch alarm health check**.
- ◆ Failover only works if the health check is genuinely measuring the thing users depend on.

ON THE EXAM

Failover configured against a private endpoint that never goes unhealthy is the classic broken DR design.

Resolver DNS Firewall

- ◆ **Filters outbound DNS queries from your VPCs** against domain lists, allowing, blocking or alerting.
- ◆ AWS-managed lists cover malware and botnet command and control. Custom lists cover your own policy.
- ◆ The point is **DNS exfiltration**: data smuggled out inside query names, where no other control looks.
- ◆ Rule groups can be bundled into a Route 53 Profile and applied across the organisation.

ON THE EXAM

Security groups and NACLs never inspect the query name. DNS-layer exfiltration is invisible to both.

Reading the scenario

THE SCENARIO SAYS

On-premises must resolve AWS private names

AWS workloads must resolve on-premises names

Same DNS config repeated across many VPCs

EU users must be served from the EU

Failover for a private, internal endpoint

Data leaving over DNS queries

IT IS POINTING AT

Inbound endpoint

Outbound endpoint plus a forwarding rule

Route 53 Profiles

Geolocation, not latency

CloudWatch alarm health check

Resolver DNS Firewall

ON THE EXAM

For every hybrid DNS question, say out loud who is asking and who holds the answer. The endpoint follows from that.

Meridian Freight runs applications in AWS that must resolve hostnames in corp.meridian.internal, held by DNS servers in the data centre. Separately, an on-premises batch system must resolve names in a private hosted zone for aws.meridian.internal. Direct Connect is already in place and routing works in both directions.

What should the solutions architect deploy?

- ☐ A An outbound endpoint with a forwarding rule for corp.meridian.internal
- ☐ B An inbound endpoint, and on-premises DNS pointed at it
- ☐ C Both an inbound and an outbound endpoint, with a forwarding rule for corp.meridian.internal
- ☐ D A private hosted zone for corp.meridian.internal associated with every VPC

◆ PAUSE AND DECIDE

What should the solutions architect deploy?

- A

 Outbound only
Solves the first requirement and leaves the batch system unable to resolve the AWS names. Half the scenario.
- B

 Inbound only
The other half. On-premises can now resolve AWS names, and AWS still cannot resolve the corporate domain.
- C

 Both endpoints
Two directions, two endpoints. The forwarding rule sends corporate queries out; the inbound endpoint gives on-premises somewhere to ask.
- D

 Copy the zone into AWS
Duplicates a zone somebody else maintains. It goes stale the moment a record changes on-premises, and silently.

THE SIGNAL

Two requirements pointing opposite ways need two endpoints. Options A and B are each exactly half the answer.

A company has 70 VPCs across 40 accounts. Every VPC needs the same three Resolver forwarding rules, two private hosted zone associations, and one DNS Firewall rule group. New VPCs appear weekly.

A platform engineer currently runs a script after each VPC is created, and it has been missed twice.

What should the solutions architect recommend?

- ☐ A Share the rules with AWS RAM and keep the script for associations
- ☐ B Build a Route 53 Profile containing all of it, share it with RAM, and associate it with each VPC
- ☐ C Add an AWS Config rule that flags VPCs missing the associations
- ☐ D Consolidate the 70 VPCs into one shared VPC

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 RAM plus the script
RAM shares the rules and does not associate them. The script, which is the thing being missed, survives.
- B

 A Route 53 Profile
One object holding zones, rules and firewall groups. Associate it once, and later additions to the Profile reach every VPC automatically.
- C

 Config rule to flag gaps
Detective. It reports the omission after the fact, which is roughly what happened both times it was missed.
- D

 One shared VPC
An enormous network restructuring in answer to a DNS configuration problem, discarding isolation the company chose deliberately.

THE SIGNAL

Option A is the pre-Profiles answer and still half right. Sharing is not associating, which is exactly the gap the script was filling.

A company operates identical application stacks in eu-west-1 and us-east-1. New data protection rules require that personal data of EU residents is processed only in the EU, with no exceptions.

The current design uses latency-based routing, and traffic analysis shows some EU users being served from us-east-1 during European network events.

What should the solutions architect change?

- ☐ A Keep latency-based routing and add health checks on the EU stack
- ☐ B Switch to geolocation routing, with a default record for unmatched queries
- ☐ C Switch to geoproximity routing with a strong positive bias toward eu-west-1
- ☐ D Use weighted routing, sending 100% of traffic to eu-west-1

◆ PAUSE AND DECIDE

What should the solutions architect change?

- A

 Latency plus health checks
Health checks change which endpoints are eligible. They do nothing about latency choosing a non-EU Region that is perfectly healthy.
- B

 Geolocation with a default
Deterministic by the user's location, which is what a legal boundary requires. The default record catches queries matching no location.
- C

 Geoproximity with bias
The closest miss. Bias shifts proportions and still permits routing outside the EU, so it cannot satisfy no exceptions.
- D

 Weighted 100%
Sends every user in the world to the EU, abandoning the US stack. It over-delivers on compliance by breaking the architecture.

THE SIGNAL

No exceptions is a phrase demanding a guarantee. Only geolocation gives one; the others express preference.

An internal application runs behind an internal Application Load Balancer in two Regions, reachable only over private addresses. The team has configured Route 53 failover records with endpoint health checks against the load balancer DNS names.

The health checks report Unhealthy permanently, and failover has never worked in testing.

What is the cause, and what should be done?

- ☐ A The health check interval is too aggressive; increase it to 30 seconds
- ☐ B Route 53 health checkers are public and cannot reach private endpoints; use CloudWatch alarm health checks
- ☐ C Add the Route 53 health checker IP ranges to the load balancer security group
- ☐ D Switch the records from failover to multivalue answer routing

◆ PAUSE AND DECIDE

What is the cause, and what should be done?

A Increase the interval

*Tuning frequency on a check that can never succeed.
Permanently unhealthy is not a flapping problem.*

C Allow the checker IPs

The sharpest distractor. Those ranges are published and this is real advice for public endpoints, and no security group rule makes a private IP routable from the internet.

B CloudWatch alarm health checks

The checkers are on the public internet and an internal ALB has no public address. An alarm on ALB metrics can be evaluated instead.

D Multivalue answer

Also health-check driven, so it inherits the identical problem and additionally abandons the active-passive design.

THE SIGNAL

Permanently unhealthy rather than intermittently means the checker never reached it at all. Ask whether it could have, before tuning anything.

An internal portal is published at portal.meridian.com. Staff in the office and in AWS should reach it over private addresses. Field engineers on the public internet must reach the same hostname through a public load balancer.

Changing the hostname for either group is not acceptable.

What should the solutions architect implement?

- A A public hosted zone and a private hosted zone, both for meridian.com, with different records
- B One public hosted zone, with the portal record pointing at the private IP
- C Two hostnames, internal and external, documented for each audience
- D A Resolver forwarding rule sending portal.meridian.com to on-premises DNS

◆ PAUSE AND DECIDE

What should the solutions architect implement?

- A** Public and private zones
Split-horizon DNS. Queries from associated VPCs hit the private zone; everyone else gets the public one. Same name, two answers.
- B** Public zone, private IP
Publishes internal addressing to the world and gives field engineers an address they cannot route to.
- C** Two hostnames
Explicitly ruled out by the scenario. It also pushes the problem onto every user and every bookmark.
- D** Forwarding rule
Forwarding changes which server answers, not what the answer is. Both audiences would still receive the same record.

THE SIGNAL

One hostname, two audiences, two answers is the split-horizon signature. The refusal to change the hostname is the planted detail.

A security review finds that a compromised instance in a private subnet has been exfiltrating data encoded inside DNS query names to an attacker-controlled domain. The subnet has no internet gateway and no NAT gateway.

Security groups, network ACLs and VPC Flow Logs were all in place and showed nothing unusual.

Which control would have detected or prevented this?

- ☐ A A stricter outbound security group on the instance
- ☐ B Route 53 Resolver DNS Firewall with a rule group blocking unapproved domains
- ☐ C AWS Network Firewall inspecting the subnet's outbound traffic
- ☐ D Enabling VPC Flow Logs at the subnet level as well

◆ PAUSE AND DECIDE

Which control would have detected or prevented this?

A Stricter security group

Security groups filter addresses and ports, never query names. The traffic went to the VPC resolver, which every instance legitimately needs.

C Network Firewall

Would inspect traffic leaving the VPC. The queries never left this way; they went to the VPC resolver, which sits outside that path.

B Resolver DNS Firewall

The only control that inspects the queried domain. It blocks or alerts on lookups to domains outside your policy.

D Subnet Flow Logs

Records the flow to the resolver as ordinary DNS. Flow logs carry addresses and ports, and no query names at all.

THE SIGNAL

No internet gateway and no NAT is the planted detail. The data left through the one path everybody permits: DNS resolution.

What to carry out of this module

- ◆ **Inbound means on-premises asks AWS. Outbound means AWS asks on-premises.** Two directions need two endpoints.
- ◆ **Sharing is not associating.** Rules and zones must be associated with each VPC, which is what Profiles automate.
- ◆ **Legal requirement means geolocation. Performance requirement means latency.** Bias is not a guarantee.
- ◆ **Health checkers are public.** Private endpoints need a CloudWatch alarm health check.
- ◆ Only DNS Firewall inspects the query name. Security groups and flow logs never do.

ON THE EXAM

Say who is asking and who holds the answer. That one sentence resolves most hybrid DNS questions before the options matter.